

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE NAME: Fundamentals of Data Science

COURSE CODE: DSA0401

COURSE OUTCOME COVERED

CO5: Design and implement end-to-end data-driven solutions by integrating pre-processing, statistical analysis, and machine learning for real-world applications (**BL3**)

Assessment Tool Weightage: 30%

QUESTIONS: 01

Total Marks:100

**Annexure A
Pseudocode Writing**

Code of Conduct:

*I A.Sai Rohit (192472144) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: A.Sai Rohit

RUBRICS FOR CALCULATION

SCENARIO BASED ASSESSMENT							
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Level	Marks
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario		
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues		
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts		
Decision Making	20%	Makes well-reasoned, logical	Makes reasonable decisions with	Makes weak decisions with	No clear decisions made or		

		decisions with strong rationale	some justification	limited justification	rationale provided		
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand		

Annexure A

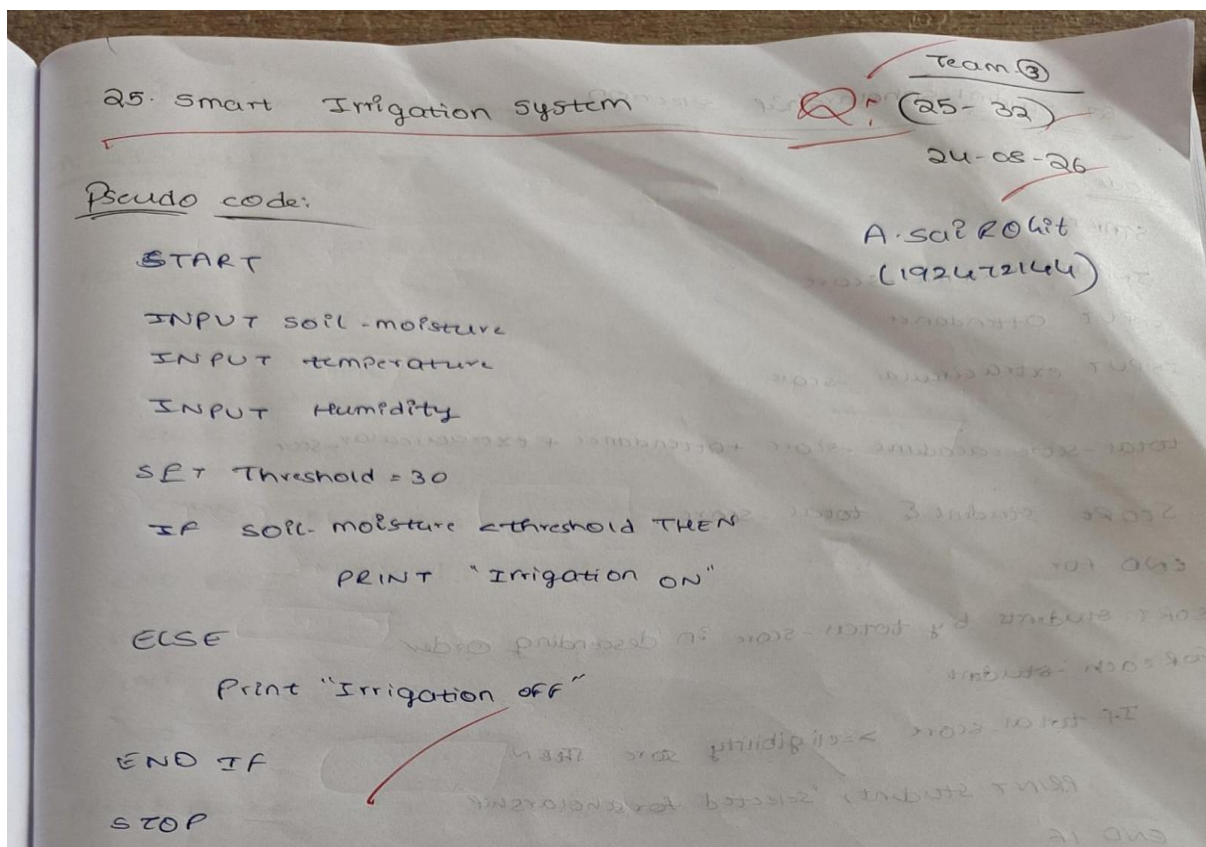
A.Sai Rohit (192472144)

Team-3 (25-32)

25. Smart Irrigation System

A farm uses sensors to collect soil moisture, temperature, and humidity data. The irrigation system should decide whether watering is required. If soil moisture falls below a threshold, irrigation should be activated automatically. Write a pseudocode for this decision-making process.

Pseudo Code:



Code:

```

import matplotlib.pyplot as plt

soil_moisture = float(input("Enter soil moisture: "))
temperature = float(input("Enter temperature: "))
humidity = float(input("Enter humidity: "))

threshold = 30

if soil_moisture < threshold:
    print("Irrigation ON")
else:
    print("Irrigation OFF")

labels = ["Soil Moisture", "Temperature", "Humidity"]
values = [soil_moisture, temperature, humidity]

plt.bar(labels, values)
plt.xlabel("Sensor Data")
plt.ylabel("Values")
plt.title("Smart Irrigation System")
plt.show()

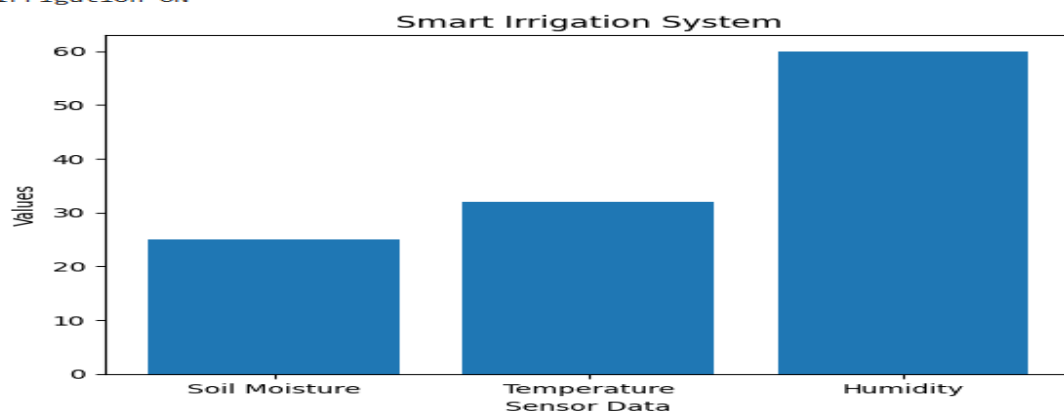
```

Output:

```

Enter soil moisture: 25
Enter temperature: 32
Enter humidity: 60
Irrigation ON

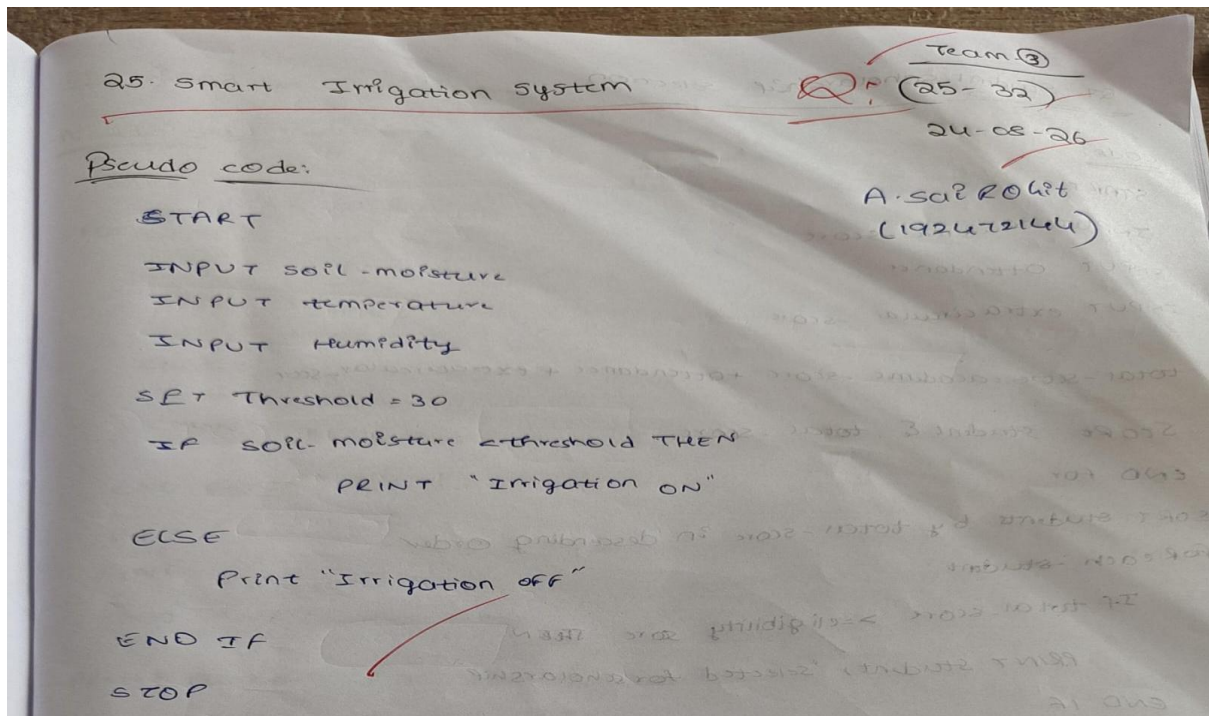
```



26. Online Shopping Cart Analysis

An e-commerce website records products added to shopping carts and completed purchases. The company wants to identify products that are frequently abandoned before checkout. The system should analyze shopping behavior and generate a report. Write a pseudocode for this analysis.

Pseudo Code:



Code:

```

import matplotlib.pyplot as plt

n = int(input("Enter number of products: "))

products = []
abandoned = []

for i in range(n):
    name = input("Enter product name: ")
    cart = int(input("Enter number of times added to cart: "))
    purchase = int(input("Enter number of completed purchases: "))

    abandoned_count = cart - purchase

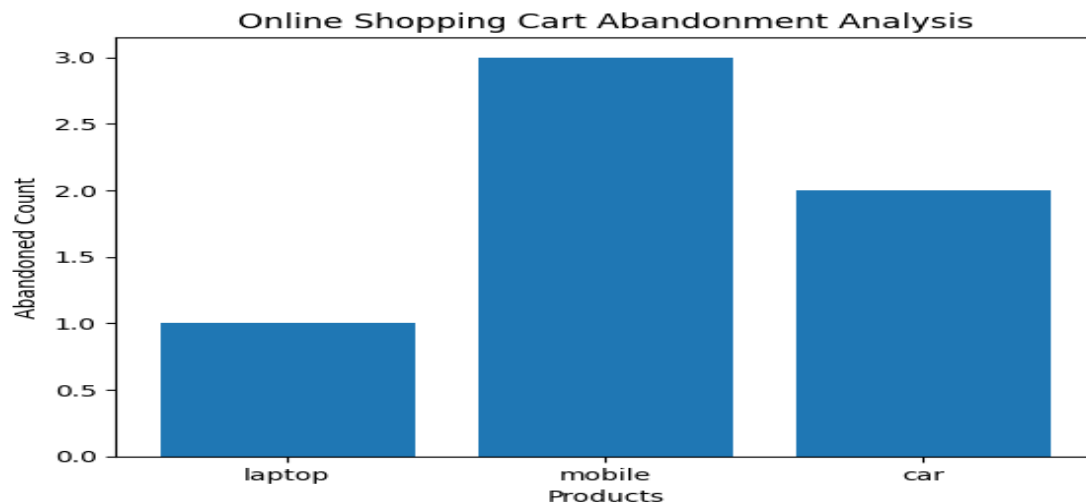
    products.append(name)
    abandoned.append(abandoned_count)

print(name, "- Abandoned:", abandoned_count)

plt.bar(products, abandoned)
plt.xlabel("Products")
plt.ylabel("Abandoned Count")
plt.title("Online Shopping Cart Abandonment Analysis")
plt.show()
  
```

Output:

```
Enter number of products: 3
Enter product name: laptop
Enter number of times added to cart: 3
Enter number of completed purchases: 2
laptop - Abandoned: 1
Enter product name: mobile
Enter number of times added to cart: 5
Enter number of completed purchases: 2
mobile - Abandoned: 3
Enter product name: car
Enter number of times added to cart: 4
Enter number of completed purchases: 2
car - Abandoned: 2
```



27. Student Scholarship Selection

A university offers scholarships based on academic performance, attendance, and extracurricular achievements. The system should evaluate all applicants and rank them according to predefined criteria. Eligible students should be selected automatically. Write a pseudocode for this selection process.

Pseudo Code:

27. Student scholarship selection

Code:

```
START
INPUT academic-score
INPUT Attendance
INPUT extracircular-score
total-score = academic-score + attendance + extracircular-score
STORE student & total-score
END FOR
SORT students by total-score in descending order
FOR each-student
    IF total-score >= eligibility score THEN
        PRINT student, "Selected for scholarship"
    END IF
END FOR
STOP
```

Code:

```
import matplotlib.pyplot as plt
```

```
n = int(input("Enter number of students: "))
```

```
names = []
```

```
totals = []
```

```
for i in range(n):
```

```
    name = input("Enter student name: ")
```

```
    academic = float(input("Enter academic score: "))
```



```

attendance = float(input("Enter attendance score: "))
extra = float(input("Enter extracurricular score: "))

total = academic + attendance + extra

names.append(name)
totals.append(total)

if total >= 150:
    print(name, "- Selected for Scholarship")
else:
    print(name, "- Not Selected")

plt.bar(names, totals)
plt.xlabel("Students")
plt.ylabel("Total Score")
plt.title("Student Scholarship Analysis")
plt.show()

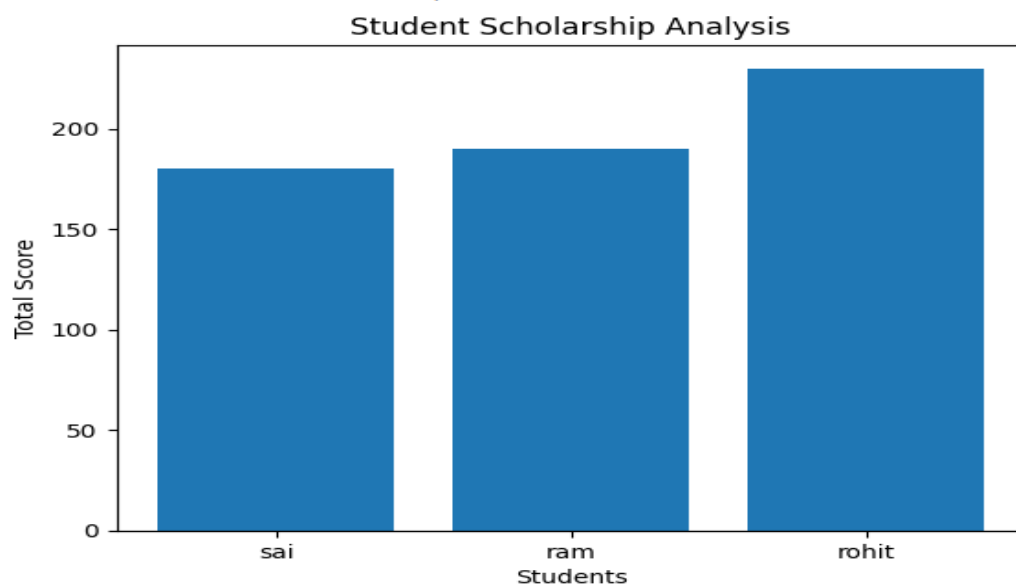
```

Output:

```

Enter number of students: 3
Enter student name: sai
Enter academic score: 80
Enter attendance score: 90
Enter extracurricular score: 10
sai - Selected for Scholarship
Enter student name: ram
Enter academic score: 90
Enter attendance score: 80
Enter extracurricular score: 20
ram - Selected for Scholarship
Enter student name: rohit
Enter academic score: 90
Enter attendance score: 90
Enter extracurricular score: 50
rohit - Selected for Scholarship

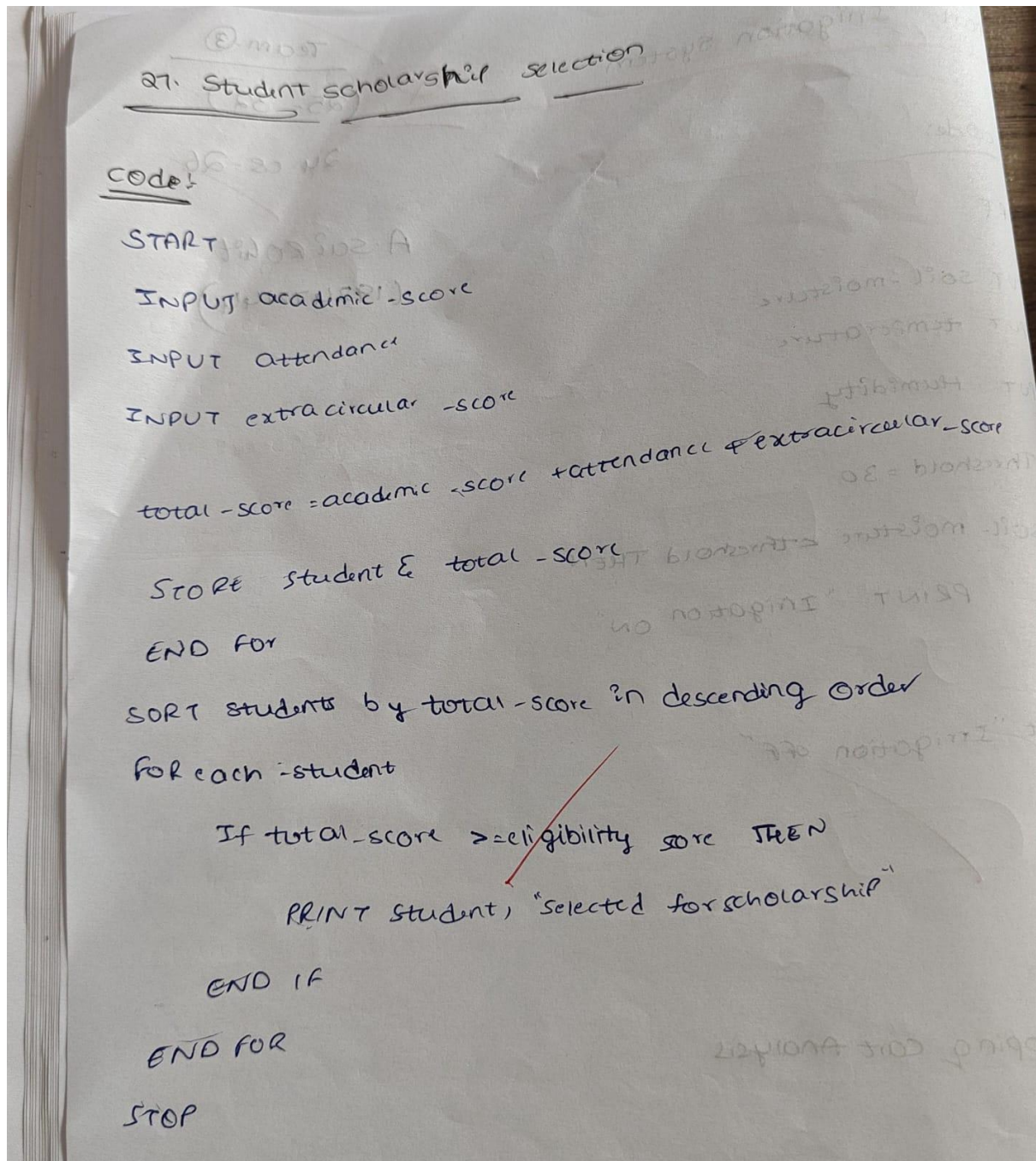
```



28. Road Accident Analysis

A transportation department maintains accident records including location, weather conditions, and time of occurrence. The department wants to identify accident-prone areas and common causes of accidents. The system should analyze historical data and generate insights. Write a pseudocode for this analysis.

Pseudo Code:



Code:

```
import matplotlib.pyplot as plt
```



```

n = int(input("Enter number of accidents: "))

locations = {}
causes = {}

for i in range(n):
    location = input("Enter accident location: ")
    cause = input("Enter accident cause: ")

    locations[location] = locations.get(location, 0) + 1
    causes[cause] = causes.get(cause, 0) + 1

print("Accidents by location:", locations)
print("Accidents by cause:", causes)

area = max(locations, key=locations.get)
cause = max(causes, key=causes.get)

print("Accident Prone Area:", area)
print("Common Cause:", cause)

plt.bar(locations.keys(), locations.values())
plt.xlabel("Location")
plt.ylabel("Number of Accidents")
plt.title("Road Accident Analysis")
plt.show()

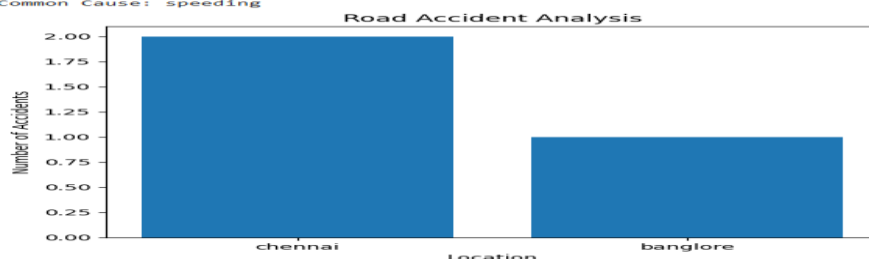
```

Output:

```

Enter number of accidents: 3
Enter accident location: chennai
Enter accident cause: speeding
Enter accident location: banglore
Enter accident cause: speeding
Enter accident location: chennai
Enter accident cause: weather
Accidents by location: {'chennai': 2, 'banglore': 1}
Accidents by cause: {'speeding': 2, 'weather': 1}
Accident Prone Area: chennai
Common Cause: speeding

```



29. Fitness Tracker Analysis

A fitness application records users' daily steps, calories burned, heart rate, and exercise duration. The app should identify whether users are meeting their fitness goals. The system should classify users as "Goal Achieved" or "Goal Not Achieved." Write a pseudocode for this analysis.

Pseudo Code:

29. Fitness tracker analysis

Pseudo code:

```

START
FOR EACH USER
  INPUT STEPS
  INPUT CALORIES
  INPUT heart-rate
  INPUT exercise-duration.
  IF STEPS >= step-goal AND
    CALORIES >= calorie-goal &
    exercise-duration >= exercise-goal THEN

```

```

    PRINT "Goal achieved"
  ELSE
    PRINT "Goal not achieved"
  END IF
END FOR
STOP.

```

30. Hotel occupancy forecasting

Code:

```
import matplotlib.pyplot as plt
```

```
steps = int(input("Enter steps: "))
```

```
calories = int(input("Enter calories burned: "))
```

```
heart_rate = int(input("Enter heart rate: "))
```

```
exercise = int(input("Enter exercise duration in minutes: "))
```

```
if steps >= 10000 and calories >= 500 and exercise >= 30:
    print("Goal Achieved")
```

```

else:
    print("Goal Not Achieved")

data = [steps, calories, heart_rate, exercise]
labels = ["Steps", "Calories", "Heart Rate", "Exercise"]

plt.bar(labels, data)
plt.xlabel("Fitness Metrics")
plt.ylabel("Values")
plt.title("Fitness Tracker Analysis")
plt.show()

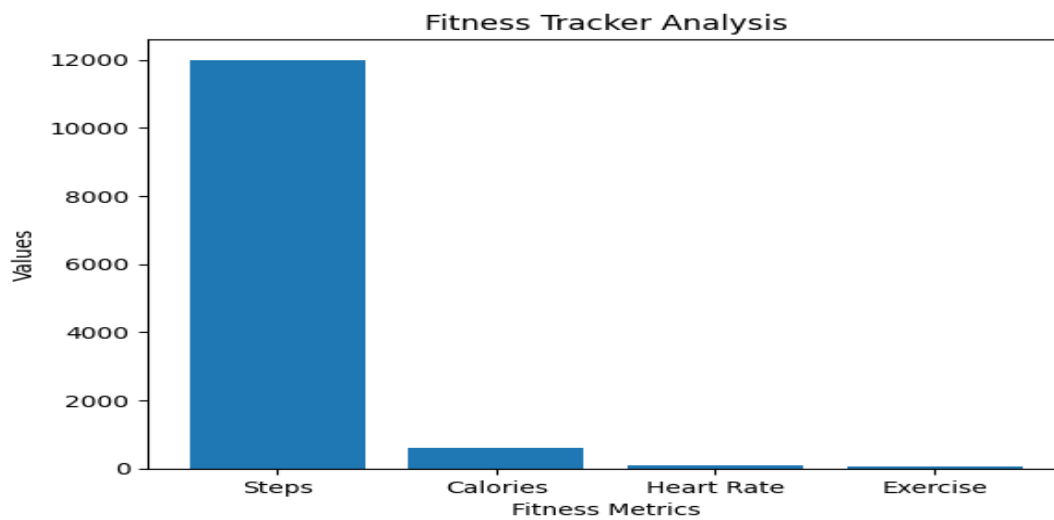
```

Output:

```

Enter steps: 12000
Enter calories burned: 600
Enter heart rate: 80
Enter exercise duration in minutes: 45
Goal Achieved

```



30. Hotel Occupancy Forecasting

A hotel chain stores historical booking data, seasonal trends, and customer preferences. The management wants to predict room occupancy for the upcoming month. The system should estimate occupancy rates and help plan staffing and resources. Write a pseudocode for this forecasting process.

Pseudo Code:

30. Hotel occupancy forecasting

Pseudo code:

```
START
INPUT historical-booking-date
INPUT seasonal-trends
INPUT customer-preferences
ANALYZE historical-booking-date
ANALYZE seasonal-trends.
CALCULATE Predicted-bookings.
Occupancy-rate =
    (Predicted bookings / (total-rooms)) * 100
PRINT Predicted-occupancy-rate
IF Occupancy-rate > 80 THEN
    PRINT "Increase staffing & resources"
ELSE
    PRINT "Normal staffing required"
END IF
STOP
```

Code:

```
import matplotlib.pyplot as plt
```

```
rooms = int(input("Enter total number of rooms: "))
```

```
bookings = int(input("Enter predicted bookings: "))
```

```
occupancy = (bookings / rooms) * 100
```

```
available = rooms - bookings
```

```
print("Predicted Occupancy Rate:", occupancy, "%")
```

```
if occupancy >= 80:
```

```
    print("High Occupancy - Increase staffing")
```

```
else:
```

```

print("Normal Occupancy")

labels = ["Booked Rooms", "Available Rooms"]
values = [bookings, available]

plt.bar(labels, values)
plt.xlabel("Room Status")
plt.ylabel("Number of Rooms")
plt.title("Hotel Occupancy Forecast")
plt.show()

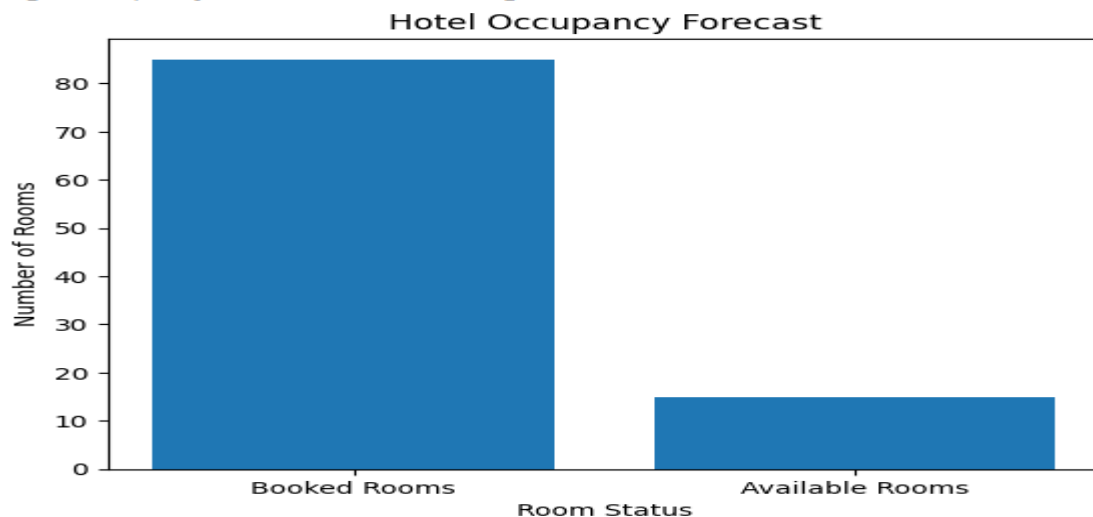
```

Output:

```

Enter total number of rooms: 100
Enter predicted bookings: 85
Predicted Occupancy Rate: 85.0 %
High Occupancy - Increase staffing

```



31. Exam Result Prediction

A school maintains records of students' attendance, assignment scores, and internal assessment marks. The administration wants to predict whether a student will pass or fail the final examination. The system should analyze historical performance data and classify students as "Pass" or "Fail." Early identification will help teachers provide additional support. Write a pseudocode for this prediction process.

Pseudo Code:

30. Hotel occupancy forecasting

Pseudo code:

START

INPUT historical-booking-date

INPUT seasonal-trends

INPUT customer-preferences

ANALYZE historical-booking-date

ANALYZE seasonal-trends.

CALCULATE predicted-bookings.

OCCUPANCY-rate =

$$\frac{\text{Predicted booking}}{\text{total-rooms}} \times 100$$

PRINT predicted-occupancy-rate

IF OCCUPANCY-rate $\geq$ 80 THEN

PRINT "Increase staffing & resources"

ELSE

PRINT "Normal staffing required"

END IF

STOP

Code:

```
import matplotlib.pyplot as plt
```

```
attendance = int(input("Enter attendance percentage: "))
```

```
assignment = int(input("Enter assignment marks: "))
```

```
internal = int(input("Enter internal marks: "))
```

```
total = attendance + assignment + internal
```

```

if total >= 100:
    print("Pass")
else:
    print("Fail")

labels = ["Attendance", "Assignment", "Internal"]
values = [attendance, assignment, internal]

plt.bar(labels, values)
plt.xlabel("Performance Factors")
plt.ylabel("Score")
plt.title("Student Exam Performance")
plt.show()

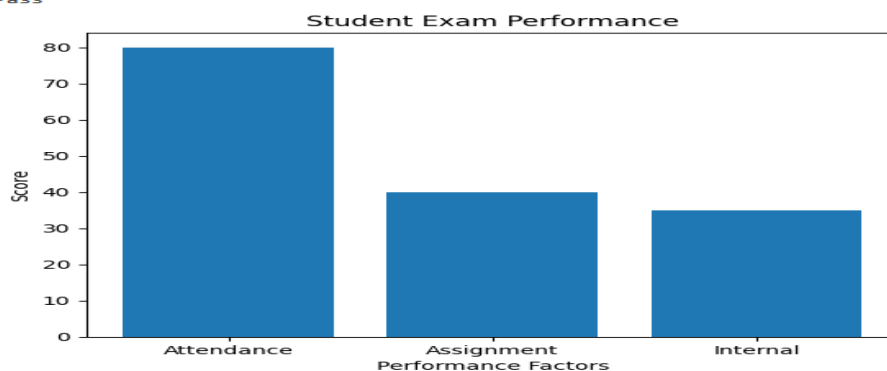
```

Output:

```

Enter attendance percentage: 80
Enter assignment marks: 40
Enter internal marks: 35
Pass

```



32. Mobile App User Retention Analysis

A mobile application tracks user login frequency, session duration, and feature usage. The company wants to identify users who are likely to stop using the app. The system should classify users as "Active" or "At Risk." The results will be used to design retention campaigns. Write a pseudocode for this analysis.

Pseudo Code:

32. Mobile App user Retention Analysis

Pseudo code:

START

FOR each user

INPUT login-frequency

INPUT session-duration

INPUT feature-usage

IF login-frequency $\geq$ min-login AND

session-duration $\geq$ minimum-duration AND

feature-usage $\geq$ minimum-usage THEN

Print user "active"

ELSE

Print user, "at risk"

END IF

END FOR

Generate retention support

STOP

Code:

```
import matplotlib.pyplot as plt
```

```
login = int(input("Enter login frequency: "))
```

```
session = int(input("Enter session duration in minutes: "))
```

```
usage = int(input("Enter feature usage count: "))

if login >= 5 and session >= 10 and usage >= 3:
    print("Active")
else:
    print("At Risk")

labels = ["Login Frequency", "Session Duration", "Feature Usage"]
values = [login, session, usage]

plt.bar(labels, values)
plt.xlabel("User Activity")
plt.ylabel("Values")
plt.title("Mobile App User Retention Analysis")
plt.show()
```

Output:

```
Enter login frequency: 8
Enter session duration in minutes: 15
Enter feature usage count: 5
Active
```

